SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	JAGADISH.V	**Reg. No.:**	192521352
Section / Batch:	B TECH IT	**Date:**	27/07/26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q57. Selection Sort – Minimum Element Reappearing Later

Apply the Brute Force technique using Selection Sort on the dataset [26, 11, 34, 11, 52, 19]. Clearly interpret the problem and justify why Selection Sort is appropriate for repeated minimum candidates. Perform the sorting step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps made. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on duplicate minimum elements.

Code of Conduct

I JAGADISH (192521352)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: JAGADISH.V

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. problem definition:

a = [26, 11, 34, 11, 52, 19]

sort the elements ascending order using selection sort.
dataset duplicate minimum value (11)

pass 1

current array:

[26, 11, 34, 11, 52, 19]

min element = 11

swap 26 <-> 11

[11, 26, 34, 11, 52, 19]

com = 5

Swaps = 1

pass 2

current array:

[11, 26, 34, 11, 52, 19]

min = 11

swap = 26 <-> 11

[11, 11, 34, 26, 52, 19]

com = 4

swap = 1

pass 3

current array:

[11, 11, 34, 26, 52, 19]

min = 19

swap 34 <-> 19

[11, 11, 19, 26, 52, 34]

com = 3

Result = 1

pass 4

current array:

[11, 11, 19, 26, 52, 34]

min = 26

[11, 11, 19, 26, 52, 34]

swap = 0

com = 2

[illegible] Result.

1. [26, 11, 34, 11, 52, 19]

2. [11, 26, 34, 11, 52, 19]

3. [11, 11, 34, 26, 52, 19]

4. [11, 11, 19, 26, 52, 34]

5. [11, 11, 19, 26, 52, 34]

$(n-1) + (n-2) + (n-3) + \dots + 1$

$5 + 4 + 3 + 2 + 1 = 15$

[illegible] Swaps = 4

$11 \leq 11$

$11 \leq 19$

$19 \leq 26$